

Patrick Amadeus Irawan

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 Website  LinkedIn  Github

Research Interests

My research goal is to build efficient, self-evolving multimodal systems that can acquire new skills and improve after its initial training. I study test-time training, inference-time scaling, continual learning, and self-refinement as mechanisms for learning from data, feedback, and interaction. My prior work spans VLM post-training and multilingual, multicultural, and low-resource multimodal evaluation.

Research Areas: efficient multimodal learning • self evolving and improving • test-time adaptation • post-training (SFT/RL) • world models • vision-language models • multimodal reasoning and evaluation

Education

Ph.D. in Natural Language Processing — MBZUAI 2025 - 2029

• Advisors: [Alham Fikri Aji](#)

Global Study Program — University of California, Davis 2023

• computational neuroscience, tech management

Bachelor in Computer Science — Institut Teknologi Bandung 2020 - 2024

• visual question answering; synthetic data generation; reasoning LLM

Work Experience

Research Engineer

Singapore

Singapore Management University

Feb 2025 – Sep 2025

- Discovered up to a 30% performance degradation in frontier VLMs upon evaluation on a multicultural visual grounding & VQA benchmark, resulting in a benchmark-focused EMNLP publication & analysis-focused preprint.
- Reduced data validation timeline by days for 50K+ spatial annotations with an automated VLM-based evaluation pipeline that automatically flagged poor polygon segmentation via foreground-color ratios.
- Minimized model guessing shortcuts and ensured benchmark robustness by engineering a heuristic-based question-generation engine to synthesize hard adversarial distractors.
- Quantified instance-specific performance decay in Video LLMs by benchmarking temporal moment segmentation in long-horizon culturally-nuanced videos.

Software Engineer

Germany

IT Bauschmiede

Nov 2024 – Jul 2025

- Automated manual enterprise operations by integrating real-time scheduling, invoicing, and geolocation systems serving 3+ scaleup companies.
- Cut system latency by ~20% while maximizing uptime and scalability by architecting a microservices migration from a legacy monolithic MVC infrastructure.

Data Science, SWE, Research Internships

Ruangguru, Biblii, Supertype

2022-2024

- Scaled command-line autograder software to support 1k+ concurrent users with 99% uptime.
- Contributed to an 8% increase in absolute approved test-case volume within 6 months at major corporation scale.
- Optimized a topic extraction pipeline, resulting in a 2x increase in speed and a 20% improvement in performance.
- Led the development of real-time AR and RAG multimodal systems for a high-stakes demo of 20k+ audiences.

Publications

* First author † Second author

[LinguDistill: Recovering Linguistic Ability in Vision Language Models via Selective Cross-Modal Distillation](#)*

Preprint 2026

VLM, SFT, post-training, knowledge distillation, architecture

Anthropogenic Regional Adaptation in Multimodal Vision-Language Model <i>VLM, SFT, post-training, data curation, evaluation, multilingual, multimodal</i>	Under Review 2026
Vision Language Models are Confused Tourists* <i>VLM, visual reasoning, robustness, hallucination, interpretability</i>	CVPR 2026
A Massive-Scale Multilingual Multi-Cultural Multimodal RAG [†] <i>RAG, multilingual, multimodal, evaluation, test-time scaling</i>	CVPR 2026
Counting to Four is Still a Chore for VLMs* <i>VLM, numerical reasoning, interpretability, RL</i>	MMFM @ CVPR 2026
Entropy2Vec: Crosslingual Language Modeling Entropy as End-to-End Learnable Language Representations* <i>LLM, multilingual, representation learning, unsupervised learning</i>	MRL @ EMNLP 2025
WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural VQA on Global Cuisines* <i>VQA, multilingual, multicultural, benchmark, evaluation, data curation</i>	NAACL 2025 (Best Theme Paper 🏆)
Seeing Culture: A Benchmark for Visual Reasoning and Grounding [†] <i>visual grounding, evaluation, object detection, multicultural</i>	EMNLP 2025
Predicting Language Model Performance on Multilingual Tasks via Proxy Models <i>multilingual, evaluation, efficient, regression, SFT</i>	NAACL 2025
Towards Efficient and Robust VQA-NLE Data Generation with Large Vision-Language Models* <i>VQA, VLM, NLP, synthetic data generation, visual reasoning</i>	COLING 2025
SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages <i>multilingual, multicultural, video understanding, speech understanding, evaluation</i>	EMNLP 2024

Achievement & Awards

OpenAI Micro Grant Award Grantee	2025
NAACL Best Theme Paper Award Winner	2025
BCG SEA Emeralds Case Competition Winner	2024
Indonesian International Student Mobility Awards @ UC Davis Awardee	2023
DOMO Higher Education Case Competition Finalist	2023
CSLeaders Full Bachelor Tuition Scholarship Awardee	2022
Gemastik XV Data Mining Division Finalist	2022

Advising & Mentorship

- **Rava Maulana** — world models and evaluation (*now Software Engineer at Tiket*)
- **Iskandar Muda** — world models and evaluation (*now M.Sc. student at Bandung Institute of Technology*)
- **Qinrong Cui** — test-time adaptation for fine-grained visual understanding (*now M.Sc. student at MBZUAI*)
- **Wilfried Ariel** — multicultural object detection and evaluation (*now Ph.D. student at NTU*)

References

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